

ECO102: Introduction to Macroeconomics

Lecture 17: Fiscal Policy & Monetary Policy (Chapter 30 & 31)

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Fiscal Policy

- The federal budget is an annual statement of the outlays and receipts of the government together with the laws and regulations that approve and support them
- The federal budget has two purposes:
 - To finance federal government programs and activities, and
 - To achieve macroeconomic objectives
- The use of the federal budget to achieve macroeconomic objectives such as full employment, sustained economic growth, and price level stability is called ***fiscal policy***.

Highlights of a Budget

- **Receipts** are the government's tax revenues, **outlays** are the government's payments, and the **deficit** is the amount by which the government's outlays exceed its receipts
- Receipts:
 - Personal income taxes
 - Social Security taxes
 - Corporate income taxes
 - Indirect taxes and other receipts
- Outlays:
 - Transfer payments
 - Expenditure on goods and services
 - Debt interest

Highlights of a Budget

- The government's budget balance is equal to receipts minus outlays

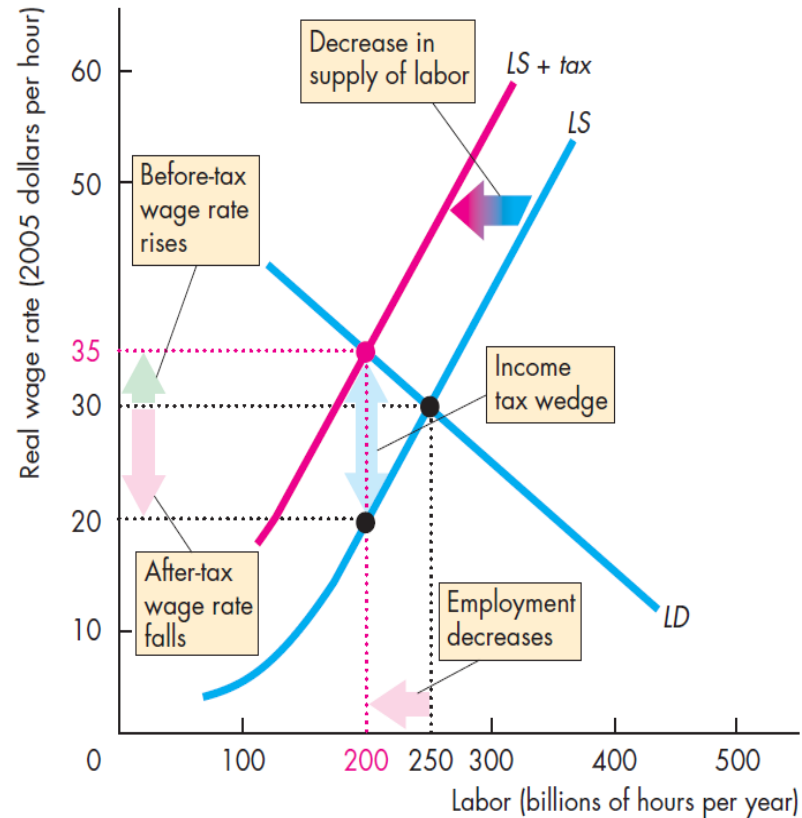
$$\text{Budget Balance} = \text{Receipts} - \text{Outlays}$$

- If Receipts = Outlays, the government has a **balanced budget**
- If Receipts > Outlays, the government has a **budget surplus**
- If Outlays > Receipts, the government has a **budget deficit**
- When the government has a budget deficit it borrows, and when it has a budget surplus it makes loan repayments.
- **Government debt** is the total amount that the government has borrowed.

Supply-Side Effects of Fiscal Policy

- The tax on labor income influences potential GDP and aggregate supply by changing the full-employment quantity of labor
- The income tax weakens the incentive to work and drives a wedge between the take-home wage of workers and the cost of labor to firms.
- The result is a smaller quantity of labor and a lower potential GDP.

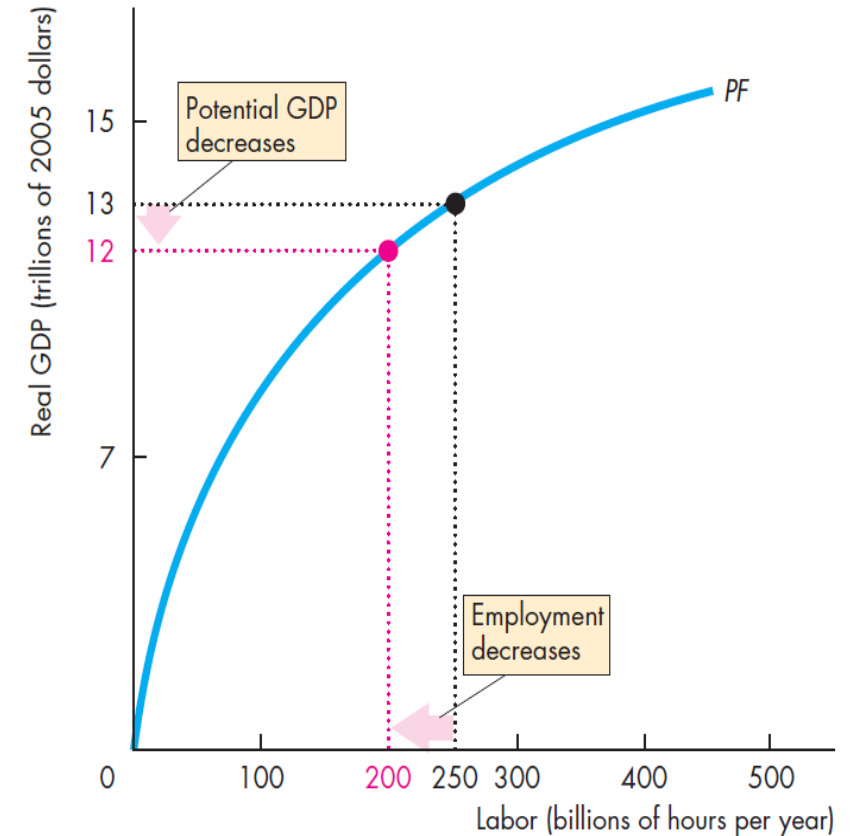
Supply-Side Effects of Fiscal Policy



(a) Income tax and the labor market

In part (a), with no income tax, the real wage rate is \$30 an hour and employment is 250 billion hours. In part (b), potential GDP is \$13 trillion.

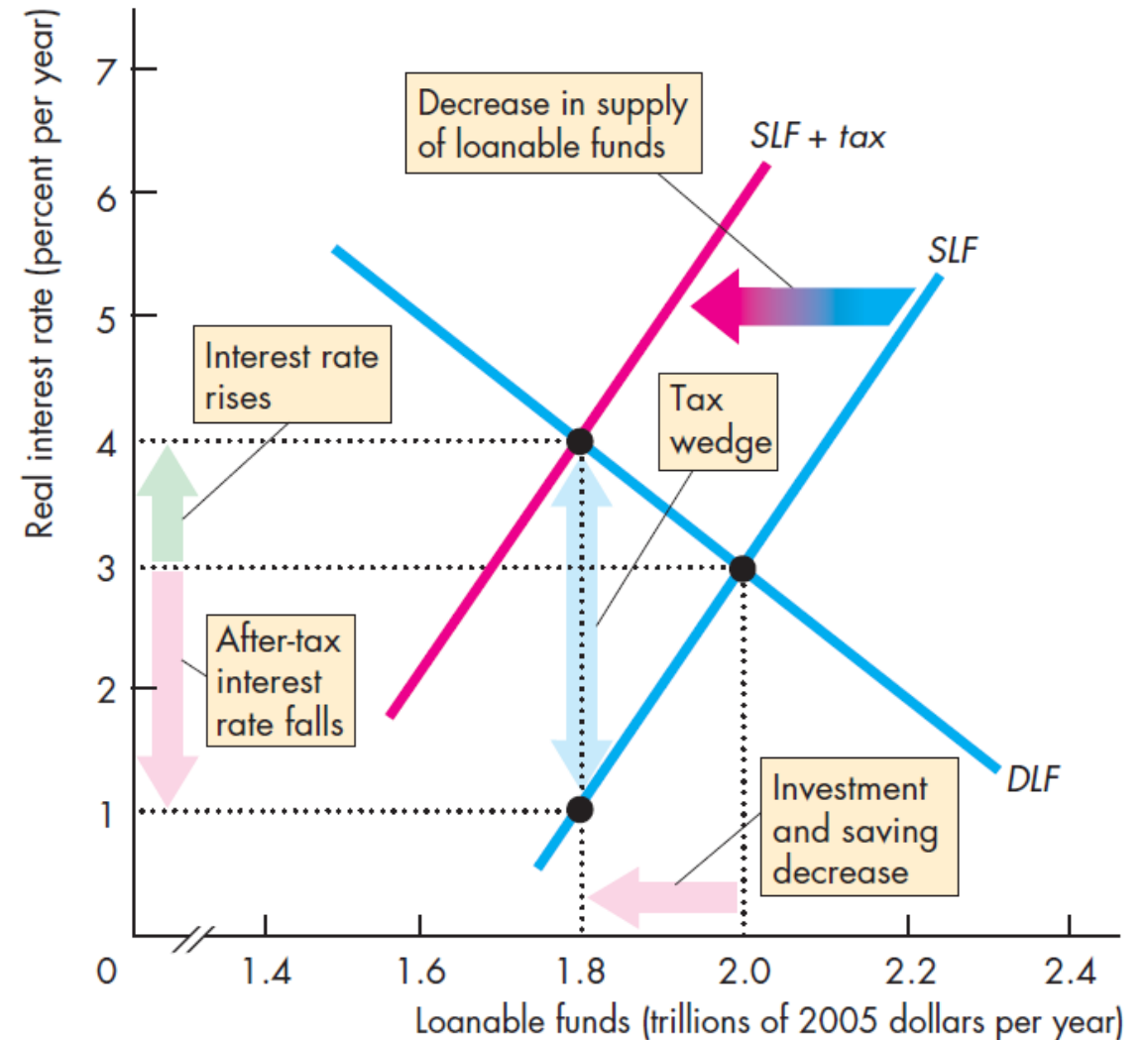
An income tax shifts the supply of labor curve leftward to $LS + tax$. The before-tax wage rate rises to \$35 an hour, the after-tax wage rate falls to \$20 an hour, and the quantity of labor employed decreases to 200 billion hours. With less labor, potential GDP decreases



(b) Income tax and potential GDP

Taxes and the Incentive to Save and Invest

With no income tax, the real interest rate is 3 percent a year and investment is \$2 trillion. An income tax shifts the supply curve leftward to $SLF + tax$. The interest rate rises to 4 percent a year, the after-tax interest rate falls to 1 percent a year, and investment decreases to \$1.8 trillion. With less investment, the real GDP growth rate decreases



Expansionary and Contractionary Fiscal Policy

- ***Expansionary Fiscal Policy:*** Refers to the government policy, when the government decides to increase government spending and/or initiates tax cuts to boost economic growth
- ***Contractionary Fiscal Policy:*** Refers to the government policy, when the government decides to decrease government spending and/or increases taxes to control inflation and close inflationary gaps
- ***THINK ABOUT HOW THE GOVERNMENT CAN USE EACH OF THESE POLICIES TO STIMULATE THE GOODS MARKET (Using the AS-AD Model)***

Monetary Policy

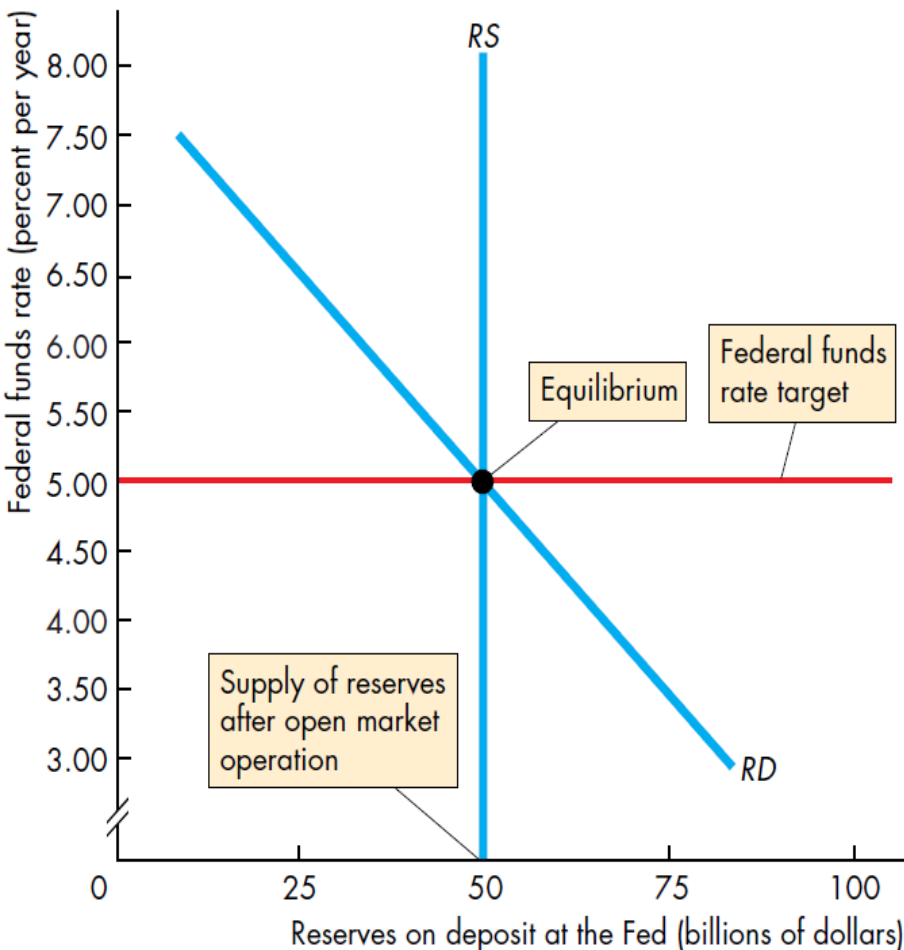
Monetary Policy

- The primary objectives of the monetary policy are:
 - Maximum employment
 - Stable Prices
 - Moderate long-term interest rates
- In the long run, these goals are in harmony and reinforce each other. But in the short run, these goals might come into conflict
- In the short run, the Central Bank faces a tradeoff between inflation and interest rates and between inflation and real GDP, employment, and unemployment
- Taking an action that is designed to lower the inflation rate and achieve stable prices might mean raising interest rates, which lowers employment and real GDP and increases the unemployment rate in the short run

Monetary Policy Instruments

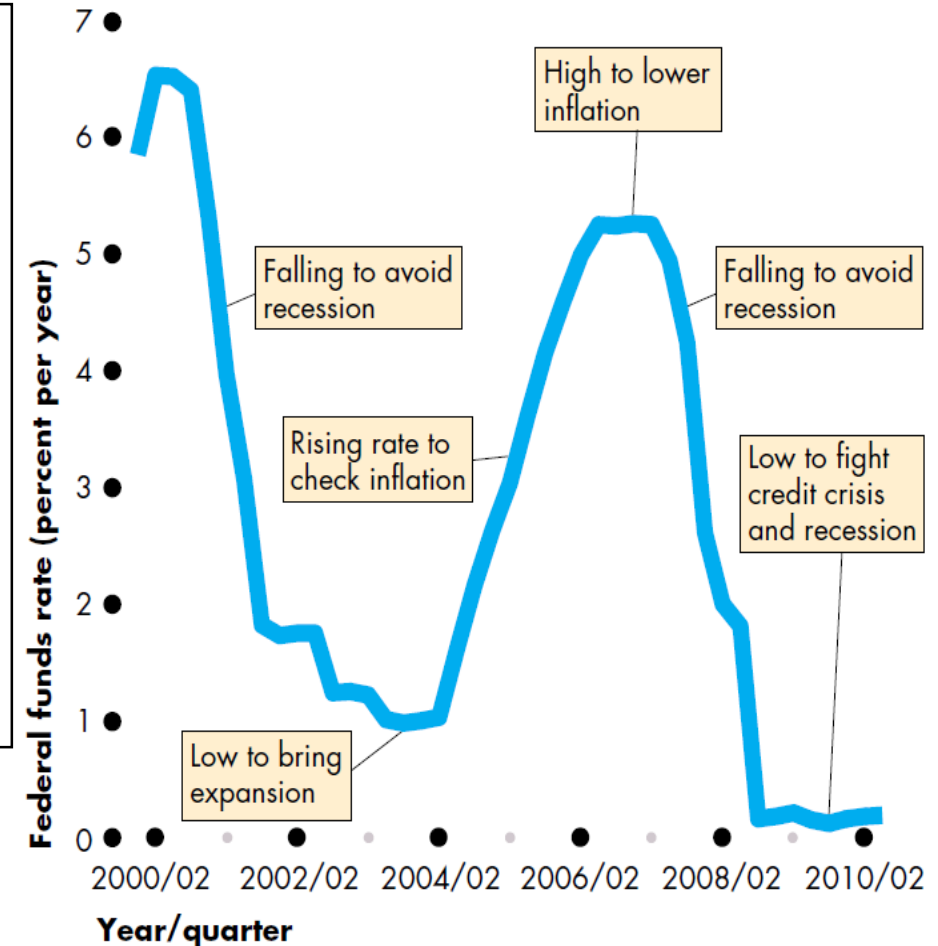
- A monetary policy instrument is a variable that the Fed can directly control or at least very closely target.
- The Fed has two possible instruments: the monetary base or the interest rate at which banks borrow and lend monetary base overnight
- The market in which the banks borrow and lend overnight is called the federal funds market and the interest rate in that market is called the ***federal funds rate***
- The Central Bank usually determines the federal funds rate through the supply of reserves, which is done through open market operations

Monetary Policy Instruments



The demand curve for reserves is RD. The quantity of reserves demanded decreases as the federal funds rate rises because the federal funds rate is the opportunity cost of holding reserves.

The supply curve of reserves is RS. The Fed uses open market operations to make the quantity of reserves supplied equal the quantity of reserves demanded at the federal funds rate target.



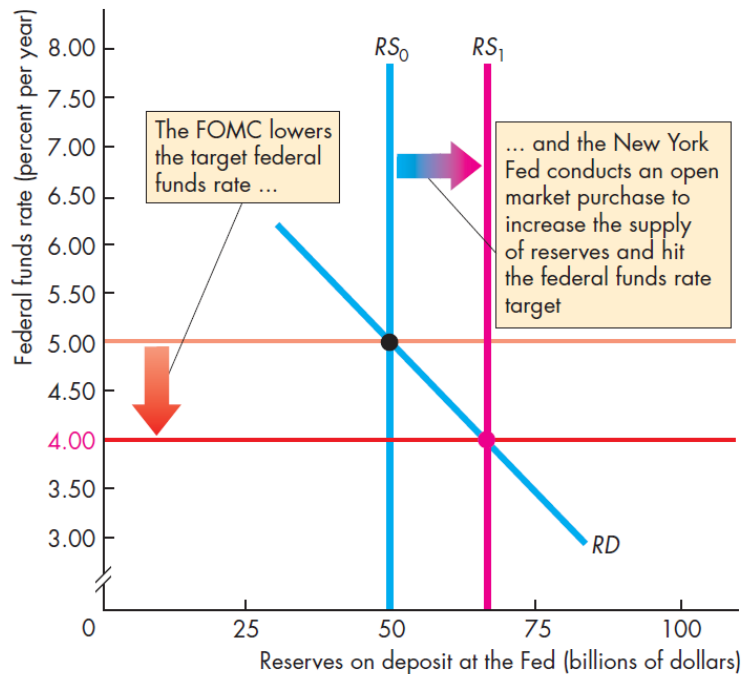
Expansionary and Contractionary Monetary Policy

- ***Expansionary Monetary Policy:*** Refers to the central bank strategy, when the bank decides to combat recession by increasing the money supply, decreasing interest rates, and loosening credit borrowing, thus increasing consumption, investments and spending, and cutting savings
- ***Contractionary Monetary Policy:*** Refers to the central bank strategy, when the bank decides to combat inflation by reducing the money supply, increasing interest rates, and tightening credit borrowing, thus reducing consumption, investments and spending, and promoting savings
- ***THINK ABOUT HOW THE CENTRAL BANK CAN USE EACH OF THESE POLICIES TO CONTROL RECESSION AND INFLATION (Goods Market, Loanable Funds Market, Money Market)***

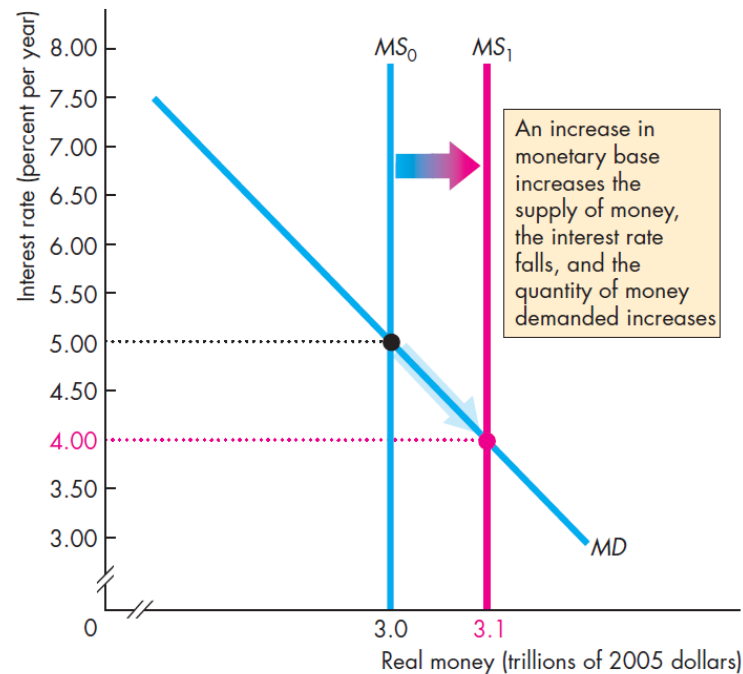
How Central Banks Fight Recessions

- If inflation is low and real GDP is below potential GDP ---> Recessionary Gap in the Economy
- To bring the economy back to the full-employment level:
 - The central bank buys securities and increases the supply of reserves of the banking system from RS_0 to RS_1 **[From graph (a)]**
 - With increased reserves, the banks create deposits by making loans and the supply of money increases. The short-term interest rate falls and the quantity of money demanded increases **[From graph (b)]**
 - Banks create money by making loans; in the short run, with a sticky price level, an increase in the supply of bank loans increases the supply of (real) loanable funds **[From graph (c)]**
 - The Aggregate Demand curve shifts to the right, closing the recessionary gap and bringing the economy back to full employment level **[Reflect on what you learnt in Chapter 27]**

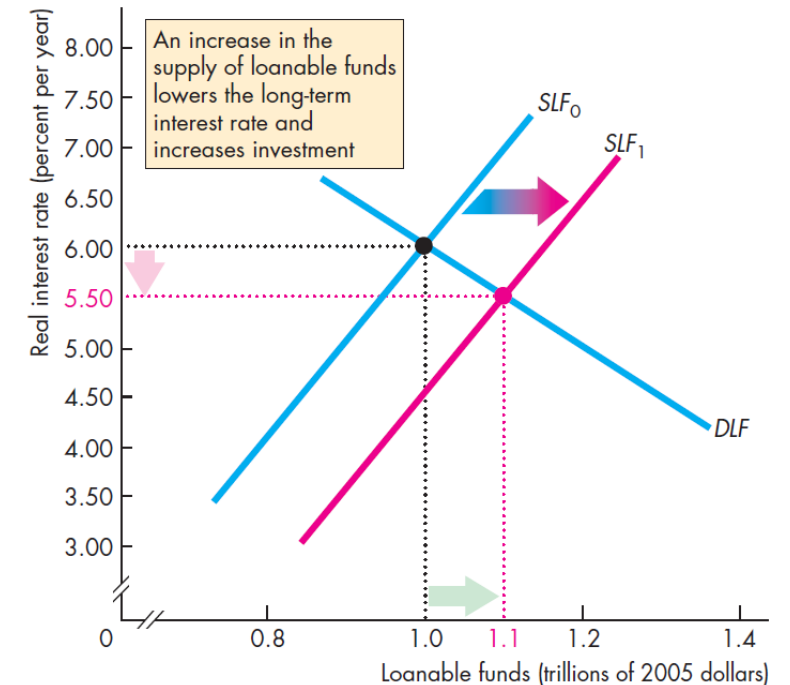
How Central Banks Fight Recessions



(a) The market for bank reserves



(b) Money market



(c) The market for loanable funds

- In (a), The Central Bank buys securities in an open market operation and increases the supply of reserves from RS_0 to RS_1 to hit the new federal funds rate target
- In (b), the supply of money increases from MS_0 to MS_1 , the short-term interest rate falls, and the quantity of money demanded increases
- In (c), an increase in the supply of bank loans increases the supply of loanable funds and shifts the supply curve from SLF_0 to SLF_1

How Central Banks Fights Inflation

- If inflation rate is too high and real GDP is above potential GDP ---> Inflationary Gap in the Economy
- To bring the economy back to the full-employment level:
 - The central bank sells securities and decreases the supply of reserves of the banking system
 - With decreased reserves, the banks shrink deposits by decreasing loans and the supply of money decreases, shifting the money supply curve leftwards. The short-term interest rate increases and the quantity of money demanded decreases
 - Banks decrease the supply of loans, shifting the supply of (real) loanable funds leftwards
 - The Aggregate Demand curve shifts to the left, closing the inflationary gap and bringing the economy back to full employment level